

AMAN GUPTA

SOFTWARE ENGINEER — FULL-STACK

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PROFESSIONAL SUMMARY

Proactive Full-Stack Software Engineer with a strong foundation in Computer Science fundamentals, modern distributed architectures, and AI/ML systems. Proven track record building high-performance web applications using **Next.js 14**, **React 18**, **TypeScript**, **Node.js**, **Express**, **PostgreSQL (Prisma)**, and **Redis**, with hands-on experience in LLM APIs (**Gemini 3.5 Flash**), semantic matching, and cloud tooling on AWS EC2.

CORE TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript, SQL

Frameworks & Web: Next.js 14, React 18, Node.js, Express, FastAPI, Prisma ORM, Tailwind CSS, REST APIs, Vite

Tools, DB & Cloud: PostgreSQL, Redis 7, Docker, AWS EC2, MongoDB Atlas, Git, GitHub, Postman, LLMs (Gemini API), Prompt Engineering, NumPy, Pandas

Engineering Strengths: Data Structures & Algorithms, Object-Oriented Programming (OOP), RESTful API Design, Database Normalization & Indexing, Problem Solving & Debugging, CI/CD & Containerization, Cross-Functional Team Collaboration

KEY ENGINEERING PROJECTS

Resume Ecosystem Builder – AI-Powered ATS & Resume Platform | Next.js 14, React 18, TypeScript, Tailwind, Prisma, PostgreSQL, Redis 7, Gemini 3.5, Docker, AWS EC2

- Architected a multi-variant resume platform with Next.js 14 App Router, TypeScript, and Tailwind CSS, featuring drag-and-drop project curation (@hello-pangea/dnd), real-time ATS keyword auditing, and instant 1-click Markdown export for LLMs.
- Engineered a hybrid ATS matching engine pairing deterministic regex keyword extraction with server-side Gemini 3.5 Flash LLM analysis, generating explainable match scores, categorized skill gaps, and in-editor bullet refinement.
- Built an atomic PostgreSQL backend via Prisma ORM with negative-index transaction reordering, backed by an in-memory Redis 7 cache with SHA-256 analysis hashing and sliding-window rate limiters, cutting query latency by 80%+.
- Containerized the full stack into an ultra-lean multi-stage Docker image (~130MB Next.js standalone) and deployed on AWS EC2 Free Tier with DuckDNS dynamic DNS synchronization and systemd boot automation.

QuickGPT – Multimodal AI Chatbot Platform | React, Node.js, Express, MongoDB, Gemini 2.5 Flash, ImageKit, Redis, Docker, AWS EC2, Nginx, Stripe, OAuth 2.0

- Architected a multimodal full-stack AI chatbot utilizing Gemini 2.5 Flash and ImageKit SDK, integrating Google OAuth 2.0 and JWTs for secure sessions and community image sharing.
- Deployed a multi-stage microservices architecture on AWS EC2 via Docker, configuring an Nginx reverse proxy with automated Let's Encrypt SSL/TLS certificates and GitHub Actions CI/CD.
- Engineered Redis-backed distributed caching, rate limiting, and token blacklisting to optimize database queries, alongside secure Stripe payment gateways for automated subscriptions.

CryptoStack – Real-Time Crypto Analytics Platform | React 18, Node.js, Express.js, Socket.io, MongoDB Atlas, Redis 7, Docker, Nginx, AWS EC2, Let's Encrypt, Systemd

- Engineered a real-time crypto analytics platform with React 18, Node, and Socket.io, delivering interactive price charts, multi-user chat, and live portfolio tracking via MongoDB Atlas.
- Architected a Redis 7 caching tier and Docker microservices pipeline, reducing API latency by 75%+ while enforcing strict limits to run the entire stack within an 82 MiB memory footprint.
- Deployed a self-healing AWS EC2 infrastructure using systemd boot automation, an Nginx reverse proxy, and Let's Encrypt SSL, achieving seamless dynamic DNS updates and zero-touch restarts.

AcadSecure – AI Plagiarism & Collusion Detection | FastAPI, React, Vite, TF-IDF, RoBERTa, DBSCAN, Ethereum, Ganache, NLTK, spaCy

- Engineered an AI-powered academic integrity engine detecting plagiarism, synthetic AI content, and student collusion rings using TF-IDF cosine similarity, fine-tuned RoBERTa transformer, and DBSCAN clustering.
- Integrated Ethereum blockchain smart contracts with Ganache to record tamper-proof audit certificates of analysis reports.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering — Global Academy of Technology, Bengaluru (2023 – 2027)
CGPA: 9.38 / 10 | Coursework in Data Structures, Algorithms, Distributed Systems, DBMS, Operating Systems, Web Technologies

Higher Secondary (Class 12) - JKBOSE in Science — Govt Boys Higher Secondary School, Jammu (2022 – 2023) | **81.4%**

CERTIFICATIONS

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- **Operating Systems Basics** — Cisco Networking Academy (November 2024)
 - **Python Programming – Basic Certification** — Python Institute / Online Fundamentals (July 2023)